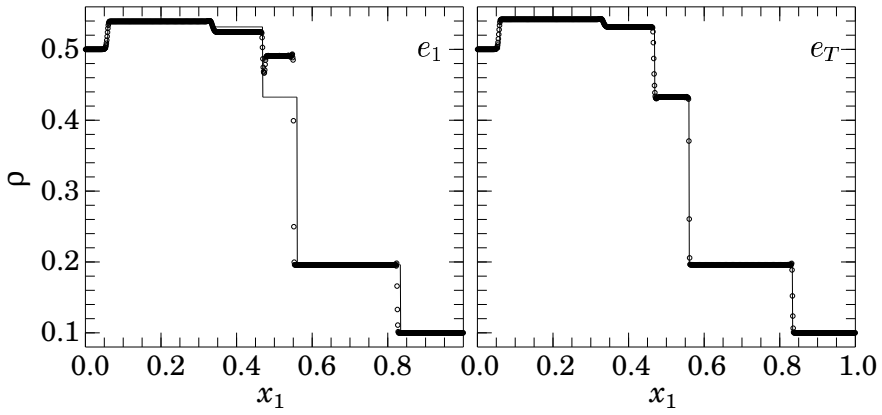
ZEUS-3D 1-D Gallery #16: Convergence and Efficiency



This is from Fig. 6 of Falle, 2002 (ApJ, 577, L123), depicting the density (ρ) from the MHD shock tube test with the left state $(\rho, v_1, v_2, v_3, B_2, B_3, p_1) = (0.5, 0, 2.0, 0, 2.5, 0, 10)$ and the right state $(0.1, -10.0, 0, 0, 2, 0, 0.1)$ with $B_1 = 2$, $\gamma = 5/3$ at time $t = 0.06$. At $t = 0$, the discontinuity is at $x_1 = 0.5$.

Open circles are the `dzeus35` solution using 1,000 zones, `CMoC`, the internal/total energy equation on the left/right (labelled e_1/e_T in the top right corner), and second-order interpolation. `dzeus35` parameters controlling the time step and artificial viscosity are: `courno=0.75`, `qcon=2.0`, and `qlin=0.2`. Lines are the results from the non-linear Riemann solver described in Ryu & Jones (1995, ApJ, 442, 228).

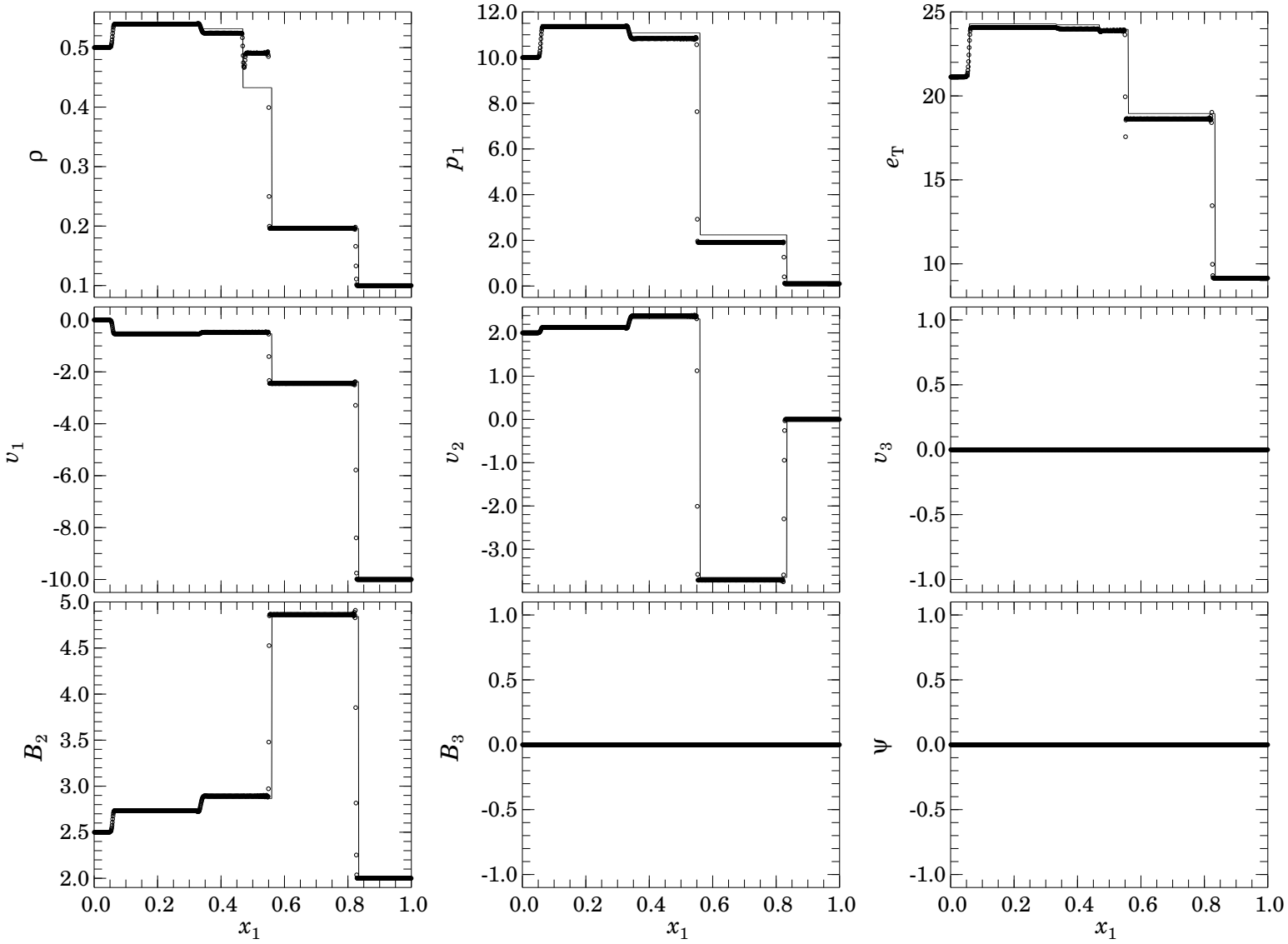
The plots show from left to right: (1) fast shock, (2) slow rarefaction (at $x_1 \sim 0.33$), (3) contact discontinuity (at $x_1 \sim 0.47$), (4) slow shock (at $x_1 \sim 0.56$), and (5) fast shock. The levels in the non-conservative solution (left) differ from those in the conservative (right) and analytical (overlays) solutions (which do agree), particularly between the contact and slow shock. Further, the wave positions in the non-conservative solution differ slightly from those in the conservative and analytical solutions.

As with Problem #15 in the 1-D Gallery, the fact that the numerical solution to the non-conservative *difference* equations differs from the analytical solution of the conservative *differential* equations comes as no surprise. In fact, the non-conservative solution shown gives the correct levels for the non-conservative problem. If a conservative solution is required, then `dzeus35` in its conservative mode should be used.

Finally, a note on efficiency. Falle's (2002) claim that *ZEUS* is 24 times slower and requires 8 times more memory than an upwinded scheme is based primarily upon the assertion that *ZEUS* requires twice as many zones to resolve the same features as an upwinded code. The results in the 1-D Gallery do not support this assertion. Certainly, if one wanted only to hold onto a standing shock, such as problem #15 of the 1-D Gallery, an upwinded scheme can do this with one zone, whereas *ZEUS*—which depends on artificial viscosity for stability—will smear the shock over several zones. But in large-scale simulations, such a scenario is irrelevant. Shocks and waves sprout spontaneously out of the dynamics moving across the grid, interacting with other waves and obstacles. In general, even fully upwinded schemes need a few zones to resolve a shock moving across a grid, as the 1-D test problems in Ryu & Jones (1995) and Falle (2002) show. For example, counting the number of zones embedded in all shocks and contacts in all twelve of Ryu & Jones' (1995) test problems (Problems 1–12 in the 1-D Gallery), one finds that in aggregate, TVD (the algorithm used by Ryu & Jones) requires ∼200 zones to resolve 34 features, whereas `dzeus35` requires ∼140 zones. Much, but not all, of this difference is attributable to the contact steepener in `dzeus35` which Ryu & Jones do not use. In Falle's (2002) Fig. 6, his upwinded scheme requires 16 zones to resolve the four discontinuities, whereas `dzeus35` requires 20 zones without the contact steepener. For the test problems in the 1-D Gallery, the claimed factor of two is simply not there and, zone for zone, *ZEUS* provides as crisp a shock-tube solution at a given resolution as most upwinded schemes.

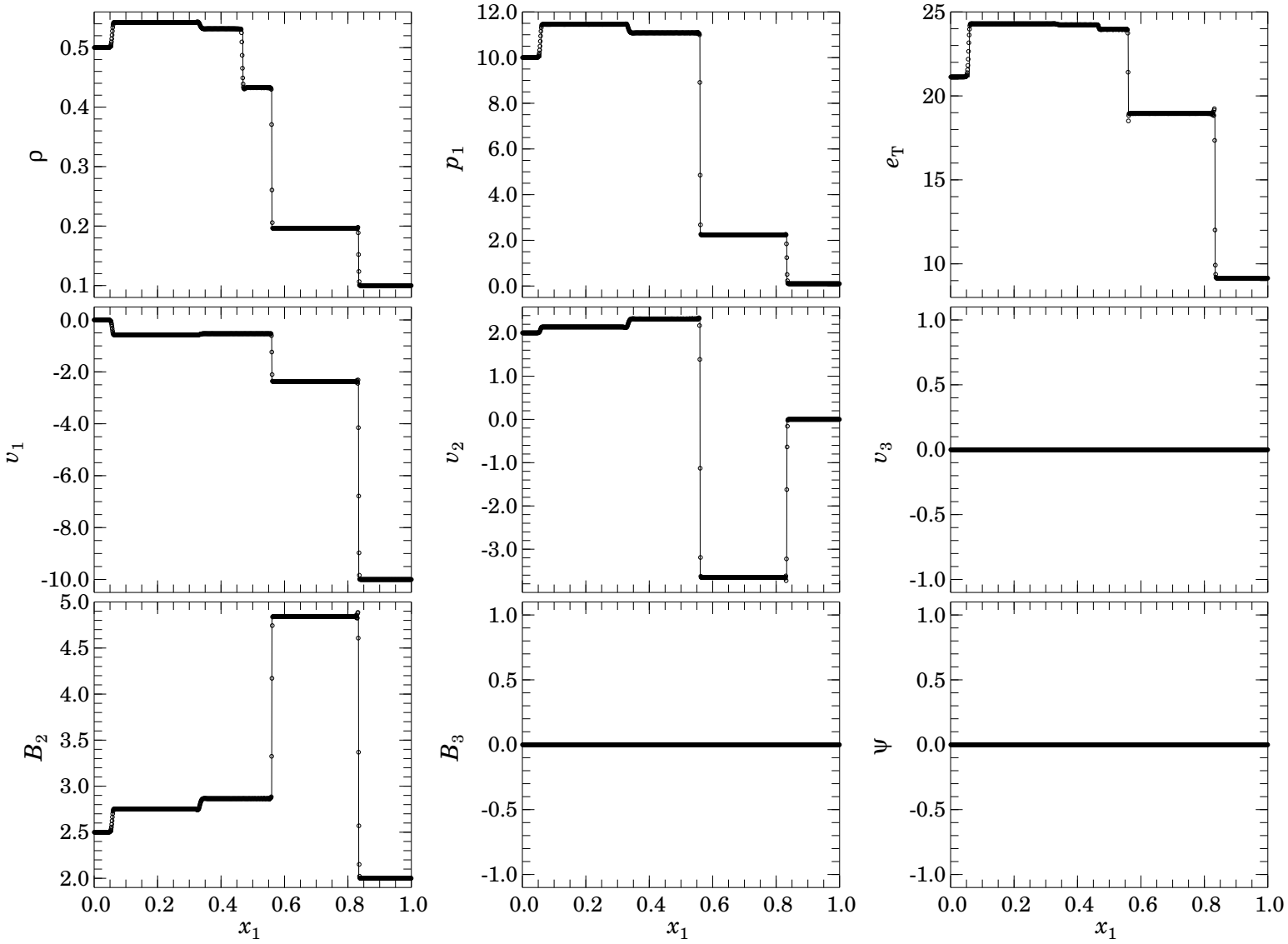
Complete non-conservative and conservative solutions are shown on the next two pages.

Non-conservative `dzeus35` solutions with conservative analytical overlays



Left state $(\rho, v_1, v_2, v_3, B_2, B_3, p_1) = (0.5, 0, 2.0, 0, 2.5, 0, 10)$ and the right state $(0.1, -10.0, 0, 0, 2, 0, 0.1)$ with $B_1 = 2$, $\gamma = 5/3$ at time $t = 0.06$. Open circles are the `dzeus35` solution using 512 zones, `CMoC`, the internal energy equation, and second-order interpolation. `dzeus35` parameters controlling the time step and artificial viscosity are: `courno=0.75`, `qcon=2.0`, and `qlin=0.2`. Lines are the results from the non-linear Riemann solver described in Ryu & Jones (1995).

Conservative `dzeus35` solutions with conservative analytical overlays



Left state $(\rho, v_1, v_2, v_3, B_2, B_3, p_1) = (0.5, 0, 2.0, 0, 2.5, 0, 10)$ and the right state $(0.1, -10.0, 0, 0, 2, 0, 0.1)$ with $B_1 = 2$, $\gamma = 5/3$ at time $t = 0.06$. Open circles are the `dzeus35` solution using 512 zones, `CMoC`, the total energy equation, and second-order interpolation. `dzeus35` parameters controlling the time step and artificial viscosity are: `courno=0.75`, `qcon=2.0`, and `qlin=0.2`. Lines are the results from the non-linear Riemann solver described in Ryu & Jones (1995).